WathiqTeam at AraGenre 2026: Class-Balanced Hybrid Retrieval with Selective LLM Reranking

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Abstract

We present WathiqTeam's system for the AraGenre 2026 Shared Task on hierarchical Arabic genre classification. The task requires assigning each Arabic text both one of six broad genres and one of 74 fine-grained specific genres organised beneath them. This setting is particularly challenging because many specific genres overlap in topic and vocabulary, while the hierarchy is highly asymmetric: the six broad genres contain substantially different numbers of specific children. We address these challenges with a broad-first retrieval architecture. Multilingual E5 compares Arabic texts with English genre definitions, an Arabic sentence encoder retrieves generated Arabic prototypes, and BM25 preserves exact lexical and structural cues. At the broad level, class-balanced pooling prevents parents with larger prototype pools from receiving more votes. After hard hierarchical gating, Weighted Reciprocal Rank Fusion combines the three independent rankings for the remaining specific labels. Qwen3 reranks only uncertain top-two broad decisions and is constrained by internal ensemble support. The submitted system achieved 0.4559 Hierarchical Macro F1, with 0.6250 Broad Macro F1 and 0.2869 Specific Macro F1. Across adaptive official submissions, class balancing and level-specific fusion produced most of the observed gain, while selective LLM reranking provided a small final improvement.

1 Introduction

Genre describes a text's communicative purpose rather than merely its subject matter, making it difficult to infer from topical words alone (Barrett et al., 2024). Arabic genre classification is further complicated by variation in register, dialect, structure, and web noise (Ferhat et al., 2024). In AraGenre 2026, these ambiguities occur inside a hierarchy of six broad and 74 specific genres (El-Haj et al., 2026). Many specific labels are semantically close, while the hierarchy is highly asymmetric: Informative contains 42 children, but Legal contains only three. This combination of class overlap and unequal support is known to make local classification less reliable (Japkowicz and Stephen, 2002; Prati et al., 2004).

Hierarchical inference offers a useful trade-off. A top-down decision reduces the fine-grained search space and guarantees a consistent parent–child output, but an incorrect parent makes the correct child unreachable (Silla and Freitas, 2011; Plaud et al., 2024). Our system therefore invests most of its robustness in the broad decision, then uses a different fusion mechanism for the more crowded specific level. This design also suits the limited supervision: only 110 labelled development texts are available for 74 leaves, whereas definitions and automatically generated prototypes can support retrieval without additional parameter fitting.

Our contributions are: (i) a multi-view retrieval formulation combining cross-lingual definitions, Arabic semantic prototypes, and lexical evidence; (ii) class-balanced broad pooling that removes the advantage of prototype-rich parents; (iii) level-specific fusion, using calibrated scores for Broad and rank fusion for Specific; and (iv) conservative LLM reranking limited to uncertain, ensemble-supported broad decisions.

2 Task and Resources

AraGenre assigns one broad label and one specific child label to each Arabic text (El-Haj et al., 2026). The official score is

$$\text{H-MF1} = \frac{\text{B-MF1} + \text{S-MF1}}{2}. \quad (1)$$

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Broad	Children	Broad ex.	Child ex.
Creative	7	12	70
Informative	42	30	420
Interactive	4	8	40
Learning	12	15	120
Legal	3	6	30
Religious	6	10	60
Total	74	81	740

Table 1: Prototype allocation. The six official broad definitions are scored separately and are not counted as broad examples.

The released development set contains 110 texts, and the hidden evaluation set contains 27,972 texts. Released resources guided initial system design and qualitative analysis. Hidden labels were never accessed, but aggregate scores from successive official submissions informed architecture and hyperparameter selection. Thus, Section 4 reports adaptive official-run comparisons rather than a fully held-out ablation. Evaluation texts were otherwise used only for automatic prediction and aggregate corpus statistics. Our corpus-level inspection of the unlabelled evaluation texts found approximate character-length percentiles of 88, 270, 662, and 1,576 at the 25th, 50th, 75th, and 90th percentiles. This wide range motivated lexical retrieval alongside dense encoders with finite context windows.

2.1 Prototype Construction

We used ChatGPT (OpenAI, 2022) to generate Arabic prototypes from the released genre definitions. Prompts requested variation in length, register, dialect, structure, topic, and realistic noise; they explicitly prohibited copying or closely paraphrasing hidden texts. We checked label and ID coverage, removed exact duplicates, and regenerated repetitive or label-inconsistent items. The exact service-side ChatGPT model version was not recorded and is therefore unavailable, which limits exact reproduction of the prototype pool.

We created 81 broad examples plus ten examples for each of the 74 specific genres (740). Broad examples were allocated by internal diversity, whereas specific labels were balanced at ten each. Table 1 exposes the resulting parent-level asymmetry: equal leaf counts still give Informative 420 child prototypes and Legal only 30. This directly motivates class-balanced rather than global neighbour aggregation.

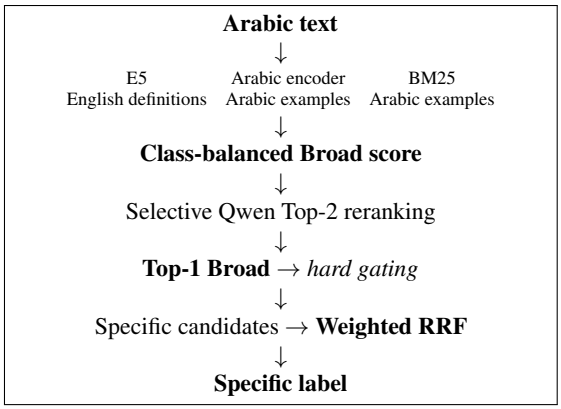


Figure 1: Final class-balanced hierarchical retrieval pipeline.

3 Final System

The submitted system treats each label as a retrieval target rather than a conventional softmax class. Sentence encoders permit efficient semantic search by comparing independently computed vectors (Reimers and Gurevych, 2019); multilingual E5 additionally supports cross-lingual matching (Wang et al., 2024). Figure 1 shows the broad-first pipeline.

3.1 Class-Balanced Broad Retrieval

Three frozen channels supply complementary evidence. Multilingual E5 matches the Arabic input to the concise English definitions. The `Arabic-Triplet-Matryoshka-V2` encoder (Nacar and Koubaa, 2024) matches it to Arabic prototypes and translated broad definitions, reducing the language and style mismatch. BM25 (Robertson and Zaragoza, 2009) preserves rare terms, formulaic openings, and exact lexical markers that dense similarity may smooth away. Scores are normalised per query and weighted 0.30, 0.50, and 0.20, respectively.

A global top-k vote would give prototype-rich parents more chances to occupy the neighbourhood. We instead fuse the three normalised scores for every prototype j and then pool an equal number of examples inside each broad class. Let $z_q(\cdot)$ denote row-wise z-normalisation for query q. The hybrid prototype score is

$$h_j = .30z_q(e_j) + .50z_q(a_j) + .20z_q(b_j), \quad (2)$$

where e_j, a_j, and b_j are E5, Arabic-encoder, and BM25 scores. For class c, we retain its five highest-scoring prototypes $T_5(c)$ and compute

$$P_c = \sum_{j \in T_5(c)} \frac{\exp(h_j/0.5)}{\sum_{\ell \in T_5(c)} \exp(h_\ell/0.5)} h_j. \quad (3)$$

Consequently, every parent receives the same retrieval budget, while several local matches reduce sensitivity to one accidental neighbour. English E5 and Arabic definition scores form $D_c = .40 z_q(d_c^{E5}) + .60 z_q(d_c^{AR})$. The final score $B_c = .85 z_q(P_c) + .15 z_q(D_c)$ gives examples most of the weight because they model linguistic realisation, while definitions anchor the intended communicative function. The selected broad label is $\arg\max_c B_c$.

3.2 Specific Retrieval and Rank Fusion

Top-1 hard gating reduces the specific decision from 74 labels to the children of the selected parent. Within that branch, E5 ranks English definitions, while the Arabic encoder and BM25 independently rank the generated examples. The top three examples per specific label are aggregated. Raw scores from dense cosine similarity and sparse lexical retrieval are not naturally calibrated; directly averaging them can make the numerically widest channel dominate. We therefore combine ranks using Weighted Reciprocal Rank Fusion (RRF) (Cormack et al., 2009):

$$S_{\mathrm{RRF}}(s) = \sum_m \frac{w_m}{10 + r_m(s)}, \quad (4)$$

where $r_m(s)$ is the rank of label s from channel m and the weights remain $\{0.30, 0.50, 0.20\}$. Each channel contributes its top 20 labels. RRF rewards agreement near the top while remaining insensitive to incompatible score ranges, which is especially useful among many overlapping fine-grained labels.

3.3 Selective LLM Reranking

LLMs can compare a small candidate set semantically, including in cross-lingual reranking (Adeyemi et al., 2024), but applying one to all 27,972 texts would increase cost and expose confident cases to prompt variation. We therefore use `Qwen3-4B-Instruct-2507` (Yang et al., 2025) only for the lowest 6% of broad-score margins, capped at 1,200 texts, and only when internal channels or example/definition branches disagree. Qwen receives the text, the top-two broad candidates, bilingual definitions, and nearby examples. A switch to the runner-up is accepted only when at least two internal channels also support it; Specific RRF is then recomputed in the revised branch. The guard makes the LLM a targeted tie-breaker rather than an unconstrained classifier.

3.4 Implementation

All evaluation encoders were frozen. E5 used query/passage prefixes and 256 tokens; the Arabic encoder used 512. Batch sizes were 64 and 32, and texts were processed in chunks of 1,024. We used random seed 20260801. Qwen was loaded in 4-bit NF4 mode with a 4,096-token input limit and at most 16 generated tokens. All outputs were produced automatically without inspecting or correcting hidden labels.

4 Final-Evaluation Experiments

All quantitative scores in this section come from official hidden-set submissions. Because aggregate feedback from earlier submissions informed later variants, Table 2 represents an adaptive system-development sequence rather than an unbiased ablation on an untouched test set. Hidden labels were never inspected. Development labels are used only for the qualitative analysis below.

4.1 Main Result and Component Analysis

The submitted system achieved 0.4559 H-MF1. Broad classification remained substantially stronger than Specific, confirming that the large, overlapping leaf space is the principal bottleneck. The gap also reflects hard-gating error propagation: an incorrect Broad decision makes the gold Specific label unavailable.

WathiqTeam ranked ninth on the official AraGenre final-evaluation leaderboard.

Table 2 traces the final-evaluation systems that led to the submitted architecture. The first four rows are alternative retrieval formulations, while the final three show the effect of class-balanced Arabic retrieval, Specific RRF, and selective Qwen. The runs are not all strict one-variable ablations: the Arabic-balanced row jointly adds an Arabic encoder, translated definitions, and balanced pooling. We therefore attribute that gain to the package as a whole, whereas the last two rows isolate level-specific RRF and selective Qwen more directly.

The largest gain is from hierarchical KNN (0.3471) to the Arabic class-balanced system (0.4419), supporting the need to correct parent pool

System	H-F1	B-F1	B-Acc	S-F1	S-Acc
E5 definition ranker	.1785	.3265	.3693	.0304	.0937
Broad-first E5	.3212	.4471	.5174	.1953	.2087
Early Broad KNN	.3438	.4812	.5805	.2065	.2141
Hierarchical KNN	.3471	.5174	.6336	.1768	.2106
Arabic balanced	.4419	.6225	.7154	.2614	.2880
+ Specific RRF	.4547	.6225	.7154	.2868	.3224
+ selective Qwen	**.4559**	**.6250**	**.7186**	**.2869**	**.3276**

Table 2: Adaptive official hidden-set submission results. B/S denote Broad/Specific Macro F1 and accuracy.

density and add Arabic-aware evidence. Replacing direct Specific score fusion with RRF retains the exact Broad output but raises Specific Macro F1 from 0.2614 to 0.2868, increasing H-MF1 to 0.4547. Selective Qwen then adds only 0.0013 H-MF1. Thus retrieval structure and level-specific fusion, not the LLM alone, explain most of the final performance.

4.2 Why Direct KNN Was Insufficient

KNN assigns a query from the labels of its nearest stored examples (Cover and Hart, 1967). This nonparametric rule was attractive because prototypes could expand the sparse 74-class supervision without retraining. However, it assumes comparable neighbourhood density across labels. In the official final-evaluation runs, adding the 740 child examples to broad retrieval improved Broad F1 from 0.4812 to 0.5174, but Specific F1 fell from 0.2065 to 0.1768: semantically close children mixed locally, and large parents supplied more neighbours. The final class-balanced pooling addresses this failure mode directly by giving every Broad class the same retrieval budget.

4.3 Qualitative Error Analysis

Because hidden labels were unavailable, we used the labelled development texts only to inspect failure modes relevant to the final architecture; no development score is mixed with the quantitative comparisons above. Motivational Writing was confused with Advice Columns when imperatives dominated the communicative intent, and Educational Explanations with Analytical Reports when formulas and formal register were shared. In one case, an unconstrained reranking prompt changed a correct definition-based Advice prediction to Motivational Writing. These observations motivated the complementary lexical/semantic channels and the agreement guard used by the final selective reranker.

5 Conclusion

Our class-balanced hierarchical retrieval system with Specific RRF and selective Qwen achieved 0.4559 official H-MF1. Across adaptive official submissions, most of the observed gain came from Arabic-aware balanced retrieval and level-specific fusion, while selective Qwen provided a smaller final improvement. Fine-grained genre overlap and hard-gating error propagation remain the main bottlenecks.

Limitations

Generated prototypes may underrepresent Arabic and inherit ChatGPT style. Hard gating cannot recover from broad errors. Weights were manually selected, and the Arabic-balanced run bundles several changes. Qwen adds cost and prompt sensitivity; the unrecorded ChatGPT version limits reproduction. Code, generation prompts, and generated prototypes are not publicly available at camera-ready submission.

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